

Chapter 6

1) Multiplexing

Defination : Multiplexing (or muxing) is a technique where multiple analog or digital signals are combined into one signal over a shared medium.

4.) Time Division Multiplexing (TDM)

Defination : Digital multiplexing technique where multiple signals share the same channel by being assigned fixed time slots.

Process :

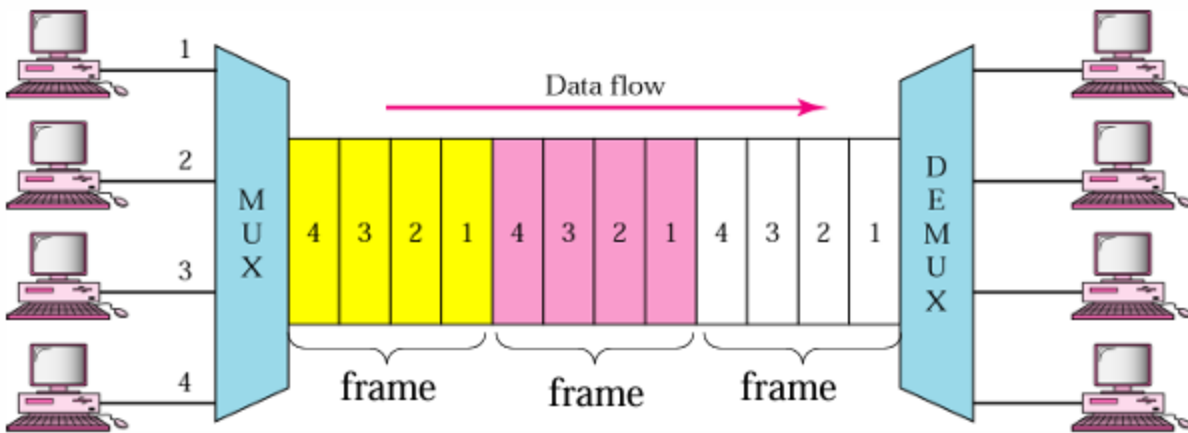
- i) Signals are divided into equal time slots.
- (ii) Multiplexer interleaves inputs into frame
- (iii) Demultiplexer reassemble signals at the receiver.

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Time-Division Multiplexing

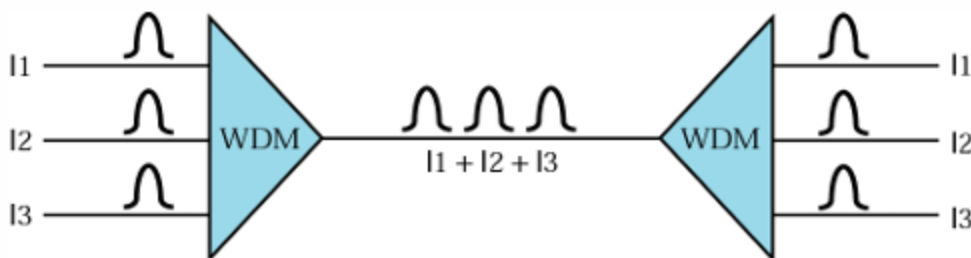


- TDM is a **digital** multiplexing technique to combine **data**.

Wavelength-division multiplexing (WDM)

Wavelength-division multiplexing (WDM) is a technology which **multiplexes** a number of **optical carrier signals** onto a **single** optical fiber by using different **wavelengths** (i.e., **colors**) of laser light. This technique enables **bidirectional** communications over one strand of fiber, as well as multiplication of capacity.

Wave Division Multiplexing (WDM)

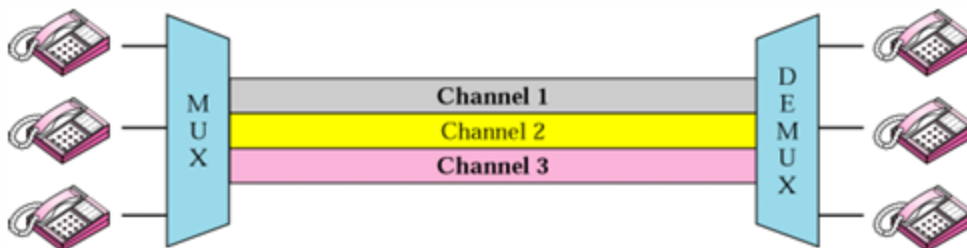


- Optic fiber provides very high data rate (50,000Gbps in theory)
- WDM is an *analog* multiplexing technique to combine *optical* signals to utilize the optical fiber
- A variation of FDM for optic channels

Frequency-division multiplexing (FDM)

Frequency-division multiplexing (FDM) is a technique by which the **total** bandwidth available in a communication medium is **divided** into a series of non-overlapping **frequency** bands, each of which is used to carry a separate signal. This allows a single transmission medium such as a **cable** or **optical** fiber to be shared by multiple independent signals. Another use is to carry separate serial bits or segments of a higher rate signal in parallel.

Frequency Division Multiplexing (FDM)



- Used when link bandwidth (hz) > the combined bandwidth of the signals to be transmitted.

TDM VS FDM (key Differences)

TDM		FDM
Feature		
Technique	Time slots allow allocated signals.	Frequency bands allocated to signals.
Signal Type	works with analog or digital	works with analog signals
Separation	framing bits (sync pulses)	Guard bands
Circuitry	Identical circuits simpler	Requires multiple carriers and bandpass filters, more complex
Efficiency	More efficient use of physical link	Less efficient (unused guard bands)
Issues	Sync errors possible.	Harmonic distortion, interference.

3/45

Multiplexing:

Math > EX-2:

100 KHz bandwidth

10 KHz Guard band between channels

5 number of channels

$$\begin{aligned}\text{Total channel bandwidth} &= 5 \times 100 \text{ KHz} \\ &= 500 \text{ KHz.}\end{aligned}$$

$$\begin{aligned}\text{Total Guard band bandwidth} &= 5 - 1 \\ &= 4 \\ &= 4 \times 10 \text{ KHz} \\ &= 40 \text{ KHz.}\end{aligned}$$

$$\begin{aligned}\text{Minimum link bandwidth} &= 500 + 40 \\ &= 540 \text{ KHz.}\end{aligned}$$

Math > Given:

Chapter 7

Networkswitching **is** the process of channeling data received from any number of input ports, to another designated port that will transmit the data to its desired destination.

Types of Switching

- There are basically three types of switching methods are available.

1. Circuit Switching.

2. Packet Switching.

3. Message Switching.

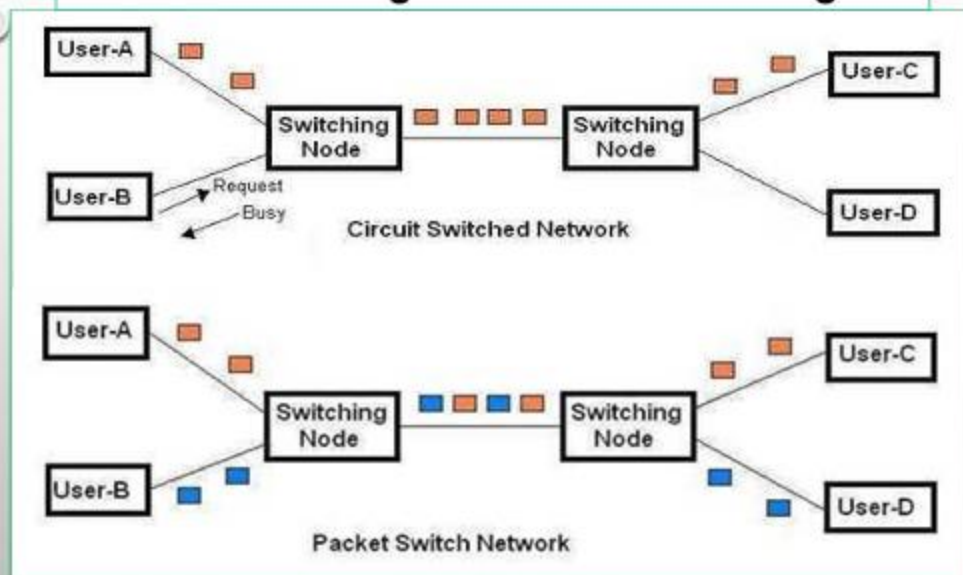
1. Circuit Switching

- **Circuit switching** is a method of implementing a telecommunications network in which two network nodes establish a dedicated communications channel (circuit) through the network before the nodes may communicate.
- The circuit guarantees the full bandwidth of the channel and remains connected for the duration of the communication session.
- The circuit functions as if the nodes were physically connected as with an electrical circuit.

2. Packet Switching

Packet-switched networks move data in separate, small blocks (packets) based on the destination address in each packet. When received, packets are reassembled in the proper sequence to make up the message.

Circuit Switching VS Packet Switching



3. Message Switching

- Message switching does not set up a dedicated channel (or circuit) between the sender and recipient during the communication session.
- In Message Switching each message is treated as an independent blocks.
- In this type of networking, each message is then transmitted from first network device to second network device through the internetwork i.e. message is transmitted from the sender to intermediary device.
- The intermediate device stores the message for a time being, after inspects it for errors, intermediate device transmitting the message to the next node with its routing information.
- Because of this reason message switching networks are called store and forward networks in networking.

3. Message Switching

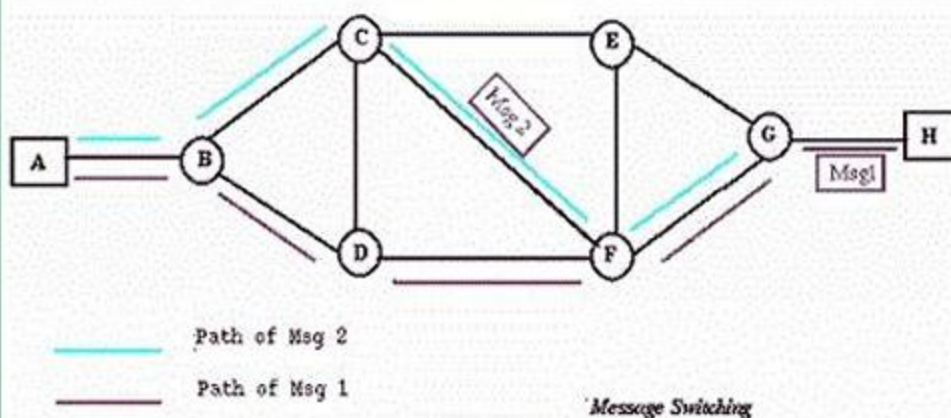


Figure: Message switching.

Chapter 8

Propagation:

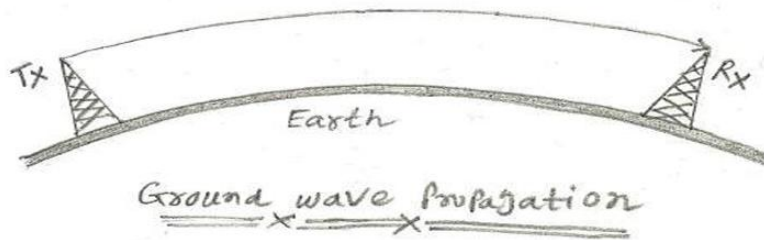
Propagation is a way in which waves travel with respect to the direction of oscillations.

Types of propagation:

1. Ground Propagation
2. Sky Propagation
3. Line of sight or Space Propagation

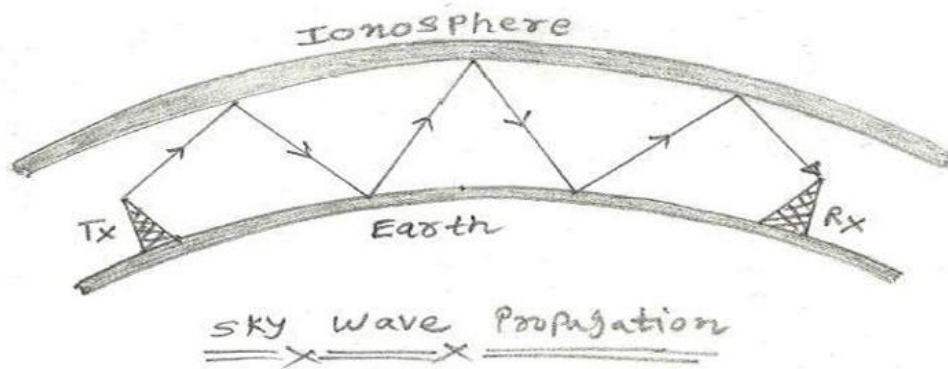
Ground Propagation:

Ground Wave propagation is a method of radio wave propagation that uses the area between the surface of the earth and the ionosphere for transmission. The ground wave can propagate a considerable distance over the earth's surface particularly in the low frequency and medium frequency portion of the radio spectrum. Ground wave radio propagation is used to provide relatively local radio communications coverage.



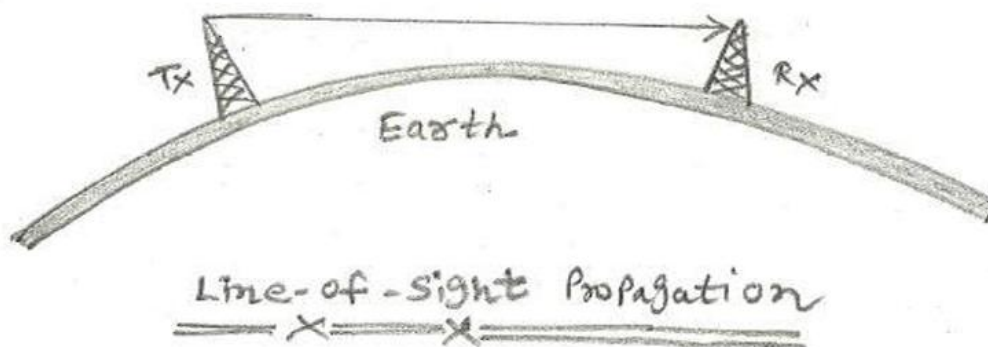
Sky Propagation:

Sky propagation refers to the propagation of radio waves that reflected or refracted back toward Earth from the ionosphere. It is mostly used in the shortwave frequency bands.



Space or Line of sight:

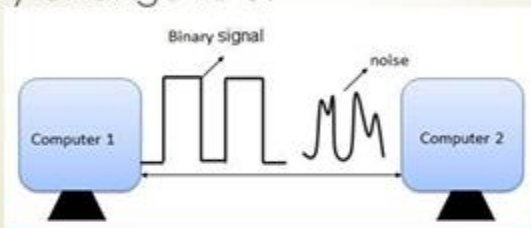
Line-of-sight propagation is a types of wave propagation that travel in a direct path from the source to the receiver. Electromagnetic transmission includes light emissions traveling in a straight line.



What Is Error?

3

- Error is a condition when the output information does not match with the input information. During transmission, digital signals suffer from noise that can introduce errors in the binary bits travelling from one system to other. That means a 0 bit may change to 1 or a 1 bit may change to 0.



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Error Detection And Correction

4

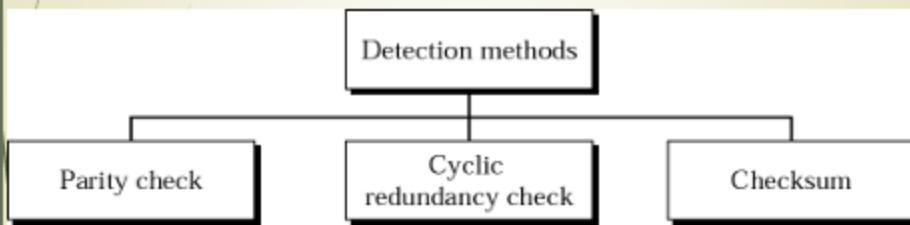
- Data can be corrupted during transmission. For reliable communication, errors must be detected and corrected Error Types.

Types of Errors:

- Single-Bit Error
 - one bit in the data unit has changed
- Burst Error
 - 2 or more bits in the data unit have changed

ERROR DETECTION

6



Error detection uses the concept of redundancy, which means adding extra bits for detecting errors at the destination.

Parity Types

12

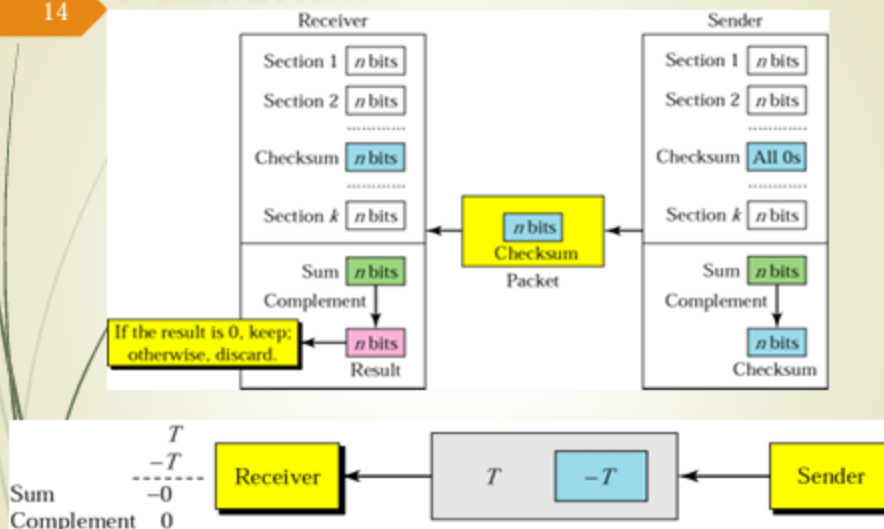
- **Even parity** -- Even parity means the number of 1's in the given word including the parity bit should be even (2,4,6,...).
- **Odd parity** -- Odd parity means the number of 1's in the given word including the parity bit should be odd (1,3,5,...).



Fig. (a)



Fig. (b)



15

Checksum: Sending

Suppose the following block of 16 bits is to be sent using a checksum of 8 bits.

10101001 00111001

The numbers are added using one's complement

	10101001
	00111001

Sum	11100010
Checksum	00011101

The pattern sent is 10101001 00111001 **00011101**

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16

Checksum: Receiving

Now suppose the receiver receives the pattern sent in Example 7 and there is no error.

10101001 00111001 00011101

When the receiver adds the three sections, it will get all 1s, which, after complementing, is all 0s and shows that there is no error.

	10101001
	00111001
	00011101
Sum	11111111
Complement	00000000 means that the pattern is OK.

CSE, DIU

⊗ Define signal Level and data level with diagram.

Signal Level: Signal level is the number of different voltage or amplitude values allowed in a signal to represent data.

Data Level: The number of symbols that represent data level is called the data level.

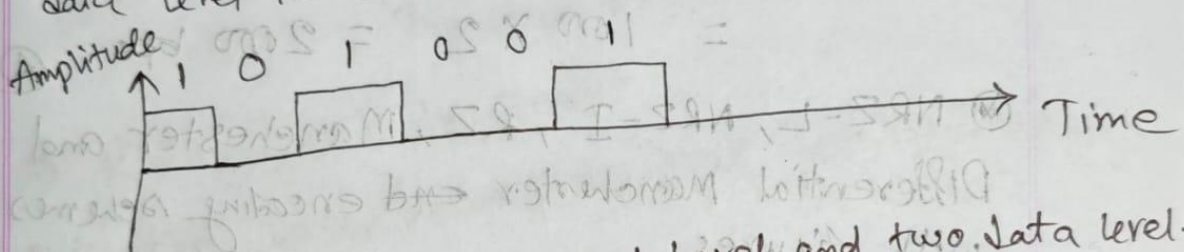


Fig: Two signal level and two data level.

Example: 2 signal level $\rightarrow +V$ and $-V$ (i)
2 data level $\rightarrow 0$ and 1 (ii)

⊗ Math from pulse rate and bit rate.

A signal has four data levels with a pulse duration of 1 ms. What is the pulse and bit rates?

Ans: Number of a data level, $L = 4$

$$\text{Pulse Duration} = 1 \text{ ms} = 1 \times 10^{-3} = 0.001 \text{ s}$$

Find the pulse rate,

$$\text{Pulse Rate} = \frac{1}{\text{Pulse Duration}}$$

$$\text{Pulse Rate} = \frac{1}{0.001} = 1000 \text{ bps}$$

Find the bit rate

$$\text{Bit} = \text{Pulse rate} \times \log_2 L$$

$$\text{Bit rate} = 1000 \times \log_2 4$$

$$= 1000 \times \log_2 2^2$$

$$= 1000 \times 2 = 2000 \text{ bps. } \textcircled{A}$$

④ NRZ-L, NRZ-I, RZ, Manchester and Differential Manchester encoding schemes with figures;

(i) NRZ (Non-Return to Zero)

(a) Non Return to Zero level (NRZ-L)

In here, positive is 0-bit and Negative is 1-bit.

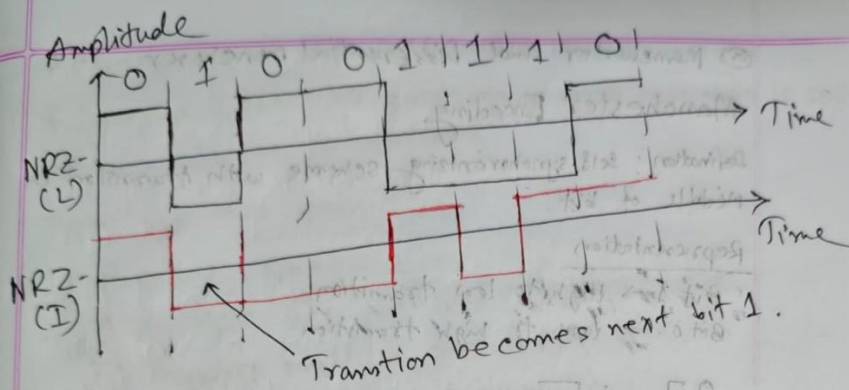
(b) Non Return to Zero - Invert (NRZ-I)

If the bit is zero it represent no change

If the bit is one it change

NRZ: Definition: Two voltage levels represent binary values.

positive voltage = 0, Negative vol = 1



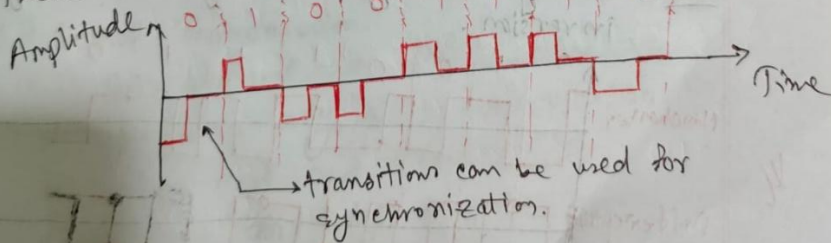
⊗ RZ (Return to zero)

Refination: Signal returns to zero between each bit.

Bit 1 → Positive to zero

Bit 0 → Negative to zero

There are 3 values in this RZ.



② Manchester and Differential Manchester

Manchester Encoding

Definition: Self synchronizing scheme with transition in middle of bit.

Representation

Bit 1 $\rightarrow$ High to low transition

Bit 0 $\rightarrow$ low to high transition

0 1

Differential Manchester

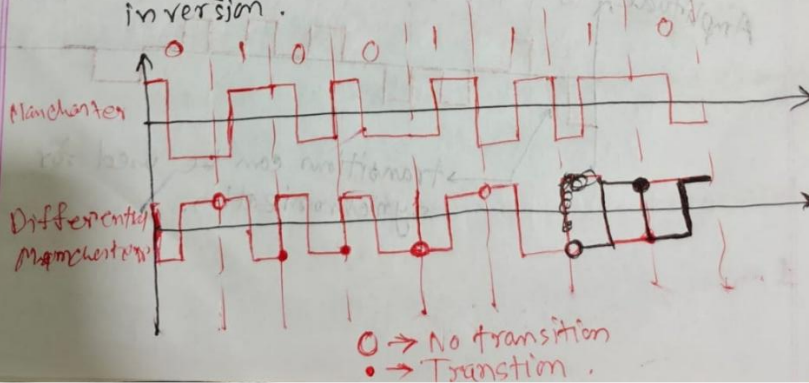
Definition: Always a transition in middle of bit interval

Representation:

i) Transition at beginning $\rightarrow$ bit = 0

ii) No transition at beginning $\rightarrow$ bit = 1

Adv: Self synchronizing, robust against polarity inversion.



④ Classification of Data Transmission Modes

⇒ Data transmission define how data is sent from one device to another.

There are two main type :

1) Parallel Transmission

- i) Multiple bits are sent simultaneously
- ii) Requires multiple connection line
- iii) High speed but expensive
- iv) Used for short distance (e.g. inside computer)

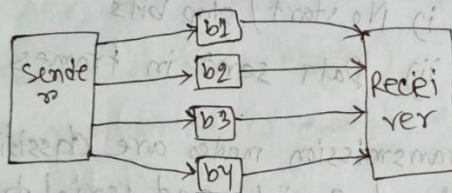


Fig: Parallel Mode

2) Serial Transmission

- i) Bits are sent one after another
- ii) Requires only one communication line.
- iii) Low cost and reliable
- iv) Used for long distance communication.

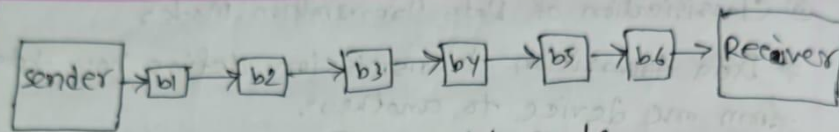


Fig: Serial mode.

Types of Serial Transmission

a) Asynchronous Transmission

- i) start bit and stop bit is used.
- ii) Data sent byte by byte

b) Synchronous Transmission

- i) No start / stop bits
- ii) Data sent in frames

Data transmission modes are classified into parallel and serial, and serial transmission is further divided into Asynchronous and Synchronous modes.

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⊗ Math from baud rate, bit rate.

Example: Baud rate = 1000 baud/s
 Bit rate = $1000 \times 4 = 4000 \text{ bps}$.

Q: If the bit rate of a signal is 3000. If each signal unit carries 6 bits. What is the baud rate?

$$\begin{aligned} \text{Baud rate} &= \text{bit rate} \div \text{signal unit bits} \\ &= 3000 \div 6 = 500 \text{ baud/s} \end{aligned}$$

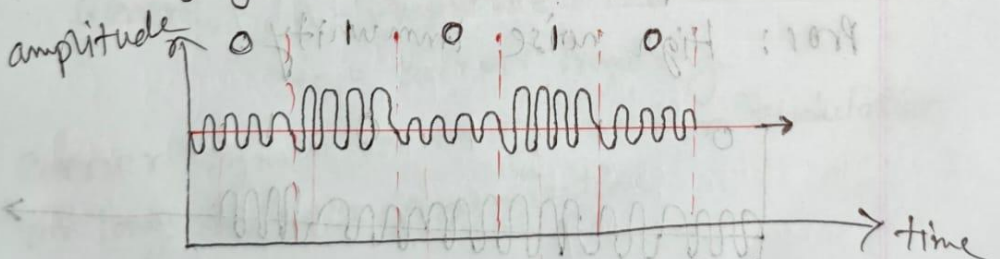
⊗ FSK, ASK, PSK, Q-PSK with appropriate figure.

ASK (Amplitude shift keying)

Binary data represented by changes in amplitude of carrier signal.

Pros: Simple, low power.

Cons: Highly noise-sensitive.



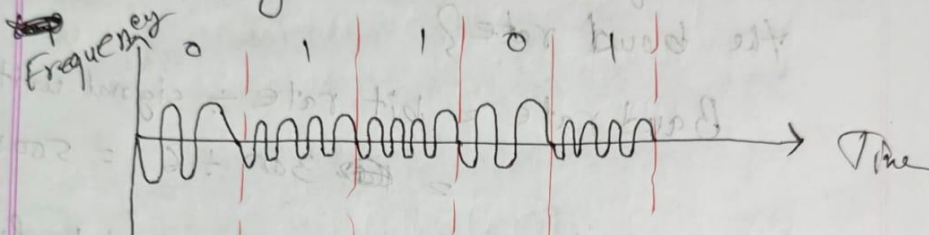
Peak amplitude during each bit duration is constant. (0 2m bit signal 10 12m signal 25 20)

FSK (Frequency Shift Keying)

Binary data represented by changes in frequency of carrier signal.

Pros: Simple, low power

Cons: Highly noise-sensitive



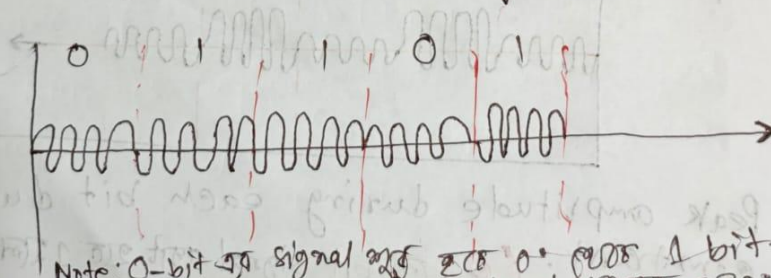
Note: $2\pi \times 200$ Hz for '0' and $2\pi \times 250$ Hz for '1'

PSK (Phase Shift Keying)

Binary data represented by phase shifts of carrier signal.

Cons: Complex demodulation

Pros: High noise immunity



Note: 0-bit is represented by $2\pi \times 200$ Hz and 1-bit is represented by $2\pi \times 250$ Hz

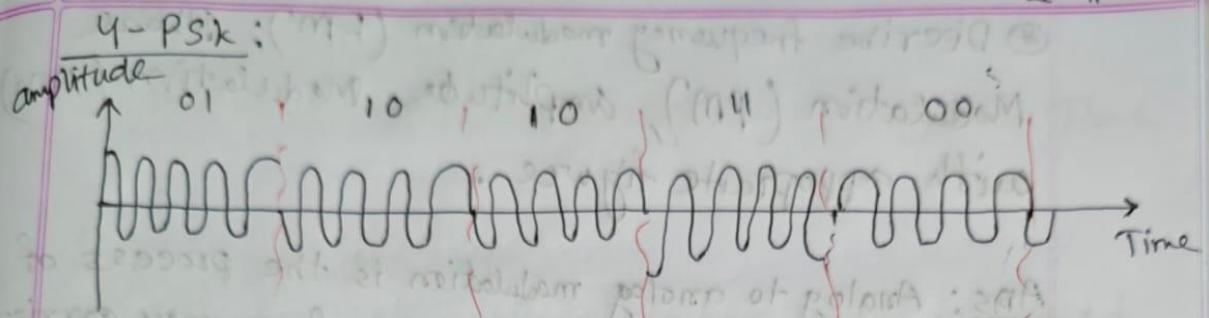


Fig: 4 PSK.

90° ଥିବା ସମୟରେ ଉଚ୍ଚ ଥିବା ଥିବା ଥିବା
 180° ବା ଉଚ୍ଚ 180° ଥିବା ଉଚ୍ଚ 0° ଥିବା ଉଚ୍ଚ
 270° ବା ଉଚ୍ଚ ଉଚ୍ଚ ଉଚ୍ଚ ଉଚ୍ଚ ଉଚ୍ଚ
 0° ବା ଉଚ୍ଚ ଉଚ୍ଚ ଉଚ୍ଚ ଉଚ୍ଚ ଉଚ୍ଚ

⊗ Carrier Signal

A high frequency sinusoidal signal that carries information from sender to receiver.

It is usually, $s(t) = A \cos(2\pi f_c t)$

Where, $A = \text{Amplitude}$

$f_c = \text{carrier frequency}$

Carrier signals are mainly used in modulation for long distance communication.

④ Describe frequency modulation (FM), Phase Modulation (PM), amplitude Modulation (AM) with appropriate figure.

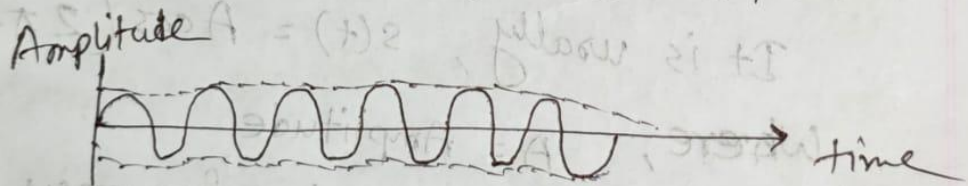
Ans: Analog to analog modulation is the process of varying a characteristic of a high frequency carrier signal according to the message signal.

There main types are:

1) Amplitude Modulation (AM)

In AM, the amplitude of the carrier signal is varied according to the instantaneous amplitude of the message signal. The frequency and phase remain constant.

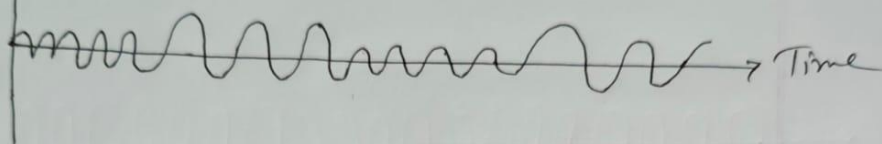
$$S(t) = (1 + k_a m(t)) \cos(2\pi f_c t)$$



Frequency Modulation (FM)

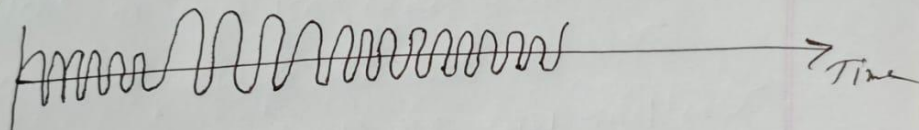
In FM, the frequency of the carrier signal is varied according to the amplitude of the message signal. Amplitude remain constant.

Frequency



Phase Modulation (PM)

IN PM, the phase of the carrier signal is varied according to the amplitude of the message signal. Amplitude remain constant.



Chapte- 6 :

Sota khatay.